Agritech Soil Rover

Master Hardware Architecture & Pinout Reference

This document serves as the canonical hardware integration guide for the 4-wheeled agritech rover. It dictates the routing of the 5200mAh LiPo power system, the timer-conflict-corrected motor driver pinouts, and actuator payload configurations.

1. Power Distribution System

Component	Input Connection	Output / Tuning Connection	Purpose
5200mAh LiPo Battery	N/A	Splits to: BTS7960s, Drill Relay, 3A Buck, 10A Buck	Main power source for the entire rover.
3A Buck Converter	LiPo `(+)` and `(-)`	**OUT (+):** Arduino VIN **OUT (-):** Arduino GND	Stepped down to **7.5V - 8.0V** to safely power the Arduino's internal regulator.
10A Buck Converter	LiPo `(+)` and `(-)`	**OUT (+):** All 5 MG996R Servos (Red) **OUT (-):** All 5 MG996R Servos (Brown) AND Arduino GND	Stepped down to **6.0V** (CV) with current (CC) fully opened. Powers high-draw servos.
Common Ground	N/A	LiPo (-), Arduino GND, 3A Buck (-), 10A Buck (-)	**CRITICAL:** Ground reference loop for logic signals.

CRITICAL COMMON GROUND RULE: All grounds across the entire rover must be tied together. If the LiPo, 10A Buck, 3A Buck, and Arduino do not share a common physical ground, control signals will float and hardware will behave erratically.

2. Motor Drivers & Drive Wheels (BTS7960)

> **Note:** The Right Motor LPWM and L-EN pins have been explicitly routed to Pins 11 and 12 to avoid the Timer 1 conflict caused by the Servo library on pins 9 and 10.

Driver	Pin / Terminal	Connection
Right BTS7960 (Timer-corrected)	**RPWM**	Arduino `Pin 6`
	LPWM	Arduino `Pin 11`
	R-EN	Arduino `Pin 10`
	L-EN	Arduino `Pin 12`
	B+ / B- (Heavy Power)	Directly to LiPo `(+)` and LiPo `(-)`
	VCC / GND (Logic 8-pin)	Arduino **5V** and Arduino **GND**
	M+ / M-	Right Wheel DC Motor
Left BTS7960	**RPWM**	Arduino `Pin 3`
	LPWM	Arduino `Pin 5`
	R-EN	Arduino `Pin 7`
	L-EN	Arduino `Pin 8`
	B+ / B- (Heavy Power)	Directly to LiPo `(+)` and LiPo `(-)`
	VCC / GND (Logic 8-pin)	Arduino **5V** and Arduino **GND**
	M+ / M-	Left Wheel DC Motor

3. Linear Actuators & Pan System (MG996R Servos)

The four actuator servos power a friction-wheel mechanism to drive the payload rods (drill and NPK sensor) vertically. These servos pull high stall currents and must strictly be powered by the 10A buck converter to prevent catastrophic damage to the Arduino logic board.

Component	Type	Power (Red/Brown)	Signal (Orange)
Drill Actuator (Left Wheel)	MG996R (360° Cont.)	10A Buck (6.0V) & GND	Arduino `Pin A2`
Drill Actuator (Right Wheel)	MG996R (360° Cont.)	10A Buck (6.0V) & GND	Arduino `Pin A3`
NPK Actuator (Left Wheel)	MG996R (360° Cont.)	10A Buck (6.0V) & GND	Arduino `Pin A4`
NPK Actuator (Right Wheel)	MG996R (360° Cont.)	10A Buck (6.0V) & GND	Arduino `Pin A5`
Camera & Sensor Pan	MG996R (180° Standard)	10A Buck (6.0V) & GND	Arduino `Pin 4`

Actuator Configuration Note: Standard 180-degree servos cannot rotate continuously to drive the linear rod mechanism. Ensure the four actuator servos are the Continuous Rotation variant. Writing 180 to these drives them forward, 0 reverses them, and 90 halts them.

4. Sensors, Relays & Payload Tools

Logic devices are safely powered by the Arduino's 5V rail (backed by the 3A buck converter). High-voltage tools like the drill motor run raw battery power switched through a dedicated relay.

Component	Pin / Terminal	Connection
Ultrasonic Sensor (HC-SR04) Mounted on Pan Servo	**Trig**	Arduino `Pin A0`
	Echo	Arduino `Pin A1`
	VCC / GND	Arduino **5V** and Arduino **GND**
Drill Motor Relay Standard 5V Relay Module	**Signal / IN**	Arduino `Pin 2`
	VCC / GND	Arduino **5V** and Arduino **GND**
	COM (Common)	Directly to LiPo `(+)`
	NO (Normally Open)	To Drill DC Motor `(+)` wire
Drill DC Motor	**Negative (-) wire**	Directly to LiPo `(-)`

Endpoint Integration: Once the Raspberry Pi 3B+ is integrated, it will establish a serial connection to the Arduino Uno via the USB port. The Arduino acts strictly as a hardware controller, executing commands passed down from the Raspberry Pi web dashboard (e.g., `DRIVE:200,200`, `DRILL_ACT:DOWN`).